

Blob Computing in Homogeneous Isotropic Distributed Systems

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Performed at
Laboratoire Interdisciplinaire des Sciences du Numériques (LISN)

Supervised by
Frédéric Gruau

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2025-26 Internship Defense

Outline

- 1 Introduction
- 2 Blob Computing Paradigm
- 3 Blobs and Fields
- 4 Language, Interpreter and GUI
- 5 Homogenization Problem
- 6 Conclusion
- 7 Resources

Context: Moore's Law Reaching Its Limits

- Current trend: transistor miniaturization increases computational power.
- Recent achievement: IBM's 0.7nm nodes (roughly 15 rows of silicon atoms).
- Challenge: approaching physical limits of miniaturization.
- Solution: scale up computationally by increasing the area of chips.

Current paradigm do not take physical distance into account, and tend not to scale well with increasing size.

LISN - Laboratoire Interdisciplinaire des Sciences du Numériques

- Located in Orsay
- Part of Université Paris-Saclay and CNRS

Supervisor: Frédéric Gruau

- Associate Professor
- Expertise: Cellular automata & Natural computing

- ① **Defined a Blob language** and implemented an interpreter
 - Reuse parallel simulation techniques from M1 internship

- ② **Developed a graphical user interface**
 - Visualize Blob program execution
 - Enable interaction with simulation

- ③ **Started designing an algorithm for the homogenization problem**
 - Related to load balancing in distributed systems
 - Self-organization using only local information
 - Still ongoing work, but preliminary results are promising

What is Blob Computing?

Definition

A massively parallel computational paradigm based on:

- Interactions of simple, locally communicating processing elements (PEs).
- Spatial arrangement of processing elements.
- Information propagation time scales linearly with distance.

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Objective

Achieve arbitrarily scalable computational power by simply increasing the number of PEs in the system

Internship Timeline

- Defense scheduled: September 2nd, 2025
- Internship completion: October 23rd, 2025
- Status: $\approx 75\%$ of time remaining after defense

Note

Information in this presentation is subject to change as work continues on the project

Blob Computing at the Micro Level

PEs

- execute the same sequence of instructions synchronously
- are arranged according to a Delaunay triangulated planar graph
- communicate only with their immediate neighbors, and cannot distinguish between neighbors of the same type
- have no knowledge of their position, nor of the graph's structure
- are anonymous
- agree on a common orientation, but not on a common direction

Computing Medium Structure

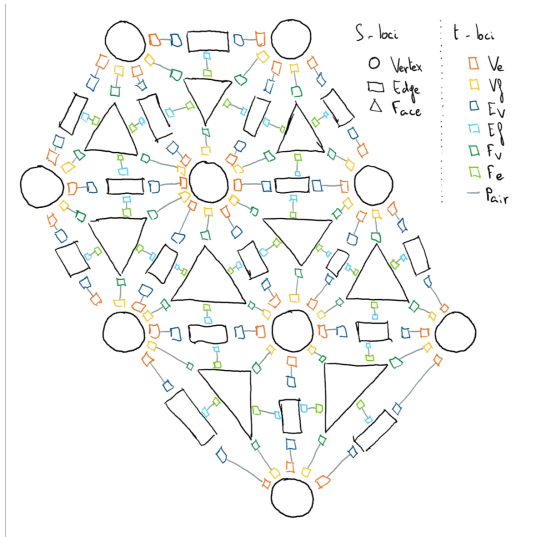
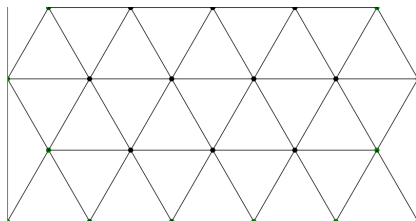
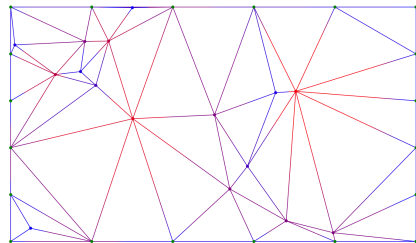


Figure: A simple example of a computing medium

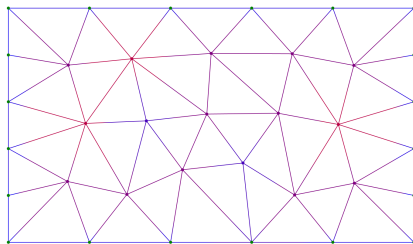
Computing Medium Structure



(a) Homogeneous but not isotropic.



(b) Isotropic but not homogeneous.



(c) Homogeneous and isotropic.

Most used medium

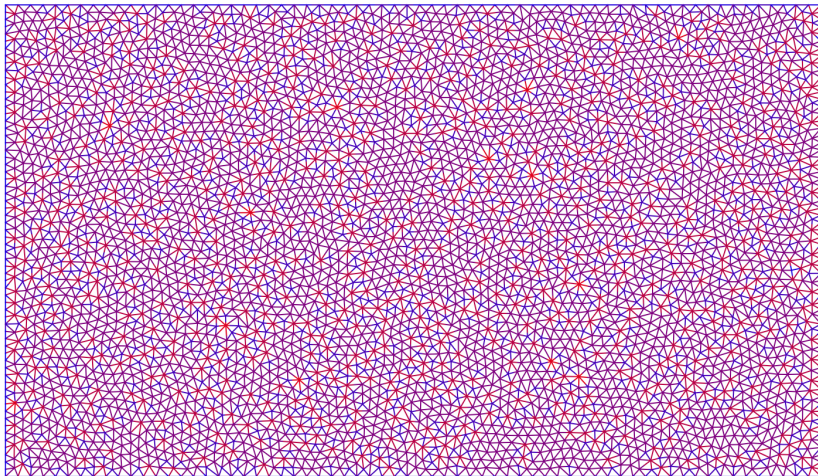


Figure: Homogeneous isotropic medium with 4355 vertices.

Blobs and Fields

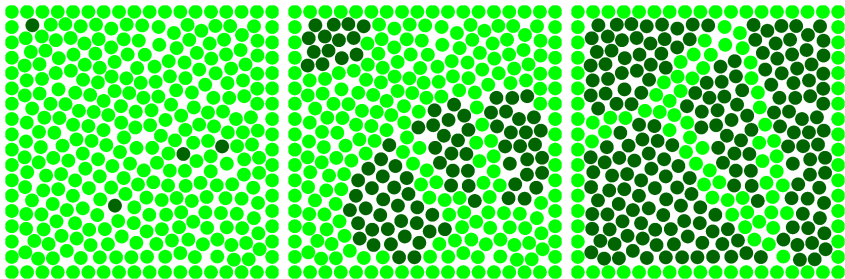


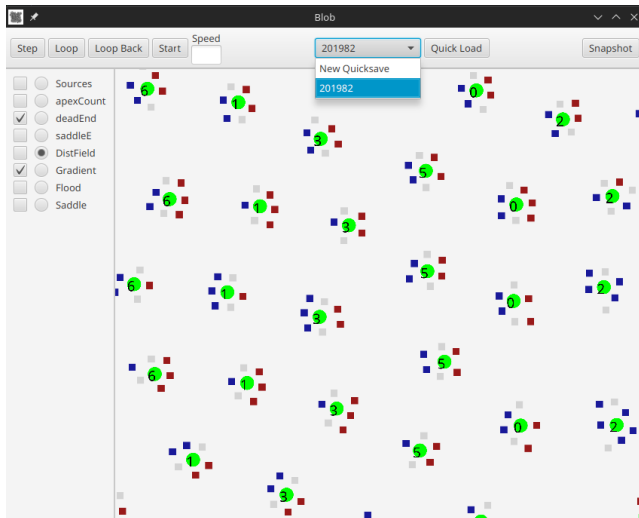
Figure: Construction of the Voronoi computation of 4 static particles in a medium.

Language: Built on top of Java to reduce development time

- Leverages Java's existing features :
 - Object-oriented programming
 - Strong typing
 - Exception handling
 - ...
- Skips the need to implement basic language features from scratch such as lexical analysis, parsing, type checking, etc.

Interesting features:

- Fully object-oriented
- Slight metaprogramming capabilities
- Many errors are caught before execution, during the construction of the instruction sequence



Homogenization Problem Definition

Definition

Homogenization: Given a set of particles distributed in a computing medium, the goal is to rearrange them so as to **maximize the minimum distance** between any two particles.

- **Practical relevance**
 - Similar to load balancing
 - Well-fitted to Blob computing due to its spatial nature
- **Research opportunity**
 - Self-organization problem
 - Rather complex but solved for the crystalline lattice

Planned Approach

- 1 Compute the distance field of the particles
- 2 Determine the Gabriel centers of the set of particles
- 3 Build the Vornoi diagram of the particles based on the Gabriel centers
- 4 Move each particle toward the center of its Vornoi cell

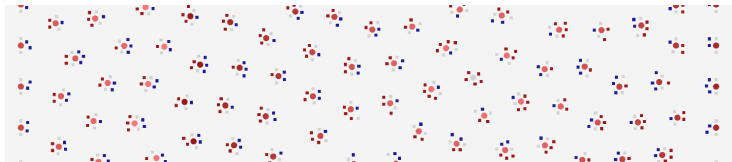
Note

Because the particles are moving, we cannot reuse the Voronoï construction of slide 12.

Distance Field

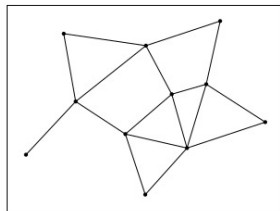


(a) Distance field.

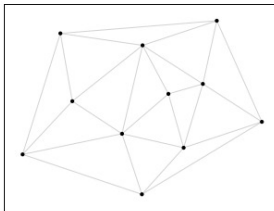


(b) Gradient field.

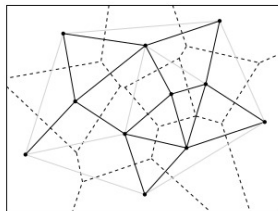
Gabriel Centers



(a) Gabriel graph



(b) Delaunay graph



(c) Both on Voronoi Diagram

Hexagonal lattice

The Gabriel centers are the vertices or edges where the gradient field descend in multiple distinct directions.

Isotropy introduces erroneous detections.

Method 1: Error Correction

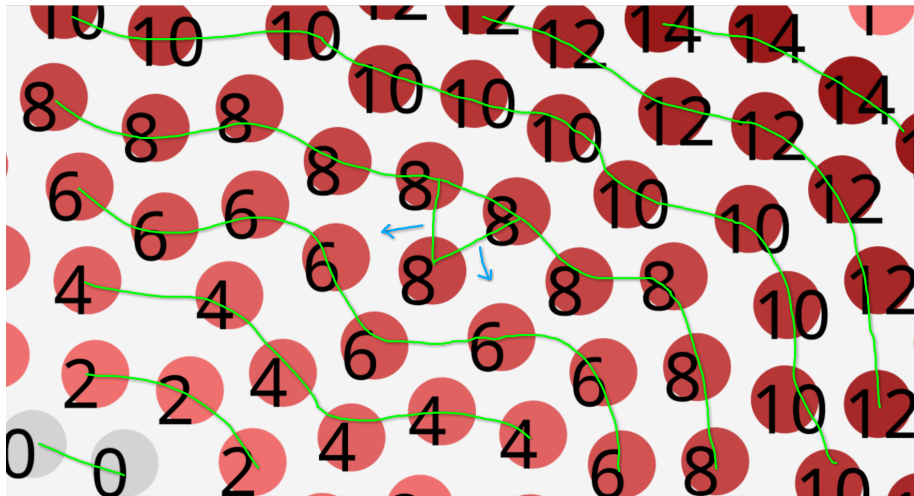
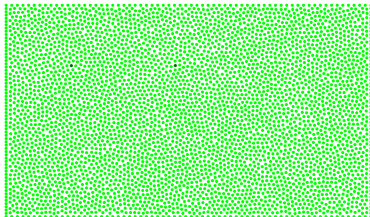
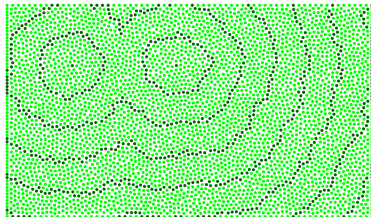


Figure: An erroneous detection.

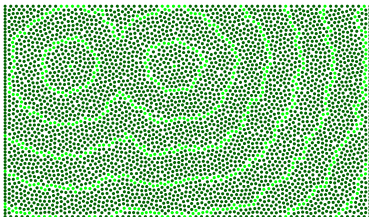
Method 2: Flooding



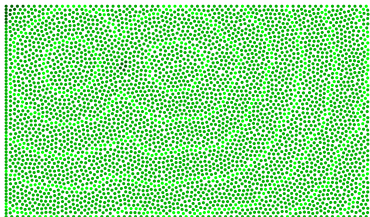
(a) The set of particles.



(b) The starting points of the floods.



(c) The flooding of the medium.



(d) The detected Gabriel centers.

Repercussions of the Work

Technical contributions:

- Blob language and interpreter
- GUI for visualizing and interacting with Blob programs
- Preliminary results on homogenization problem

Personal impact:

- Gained experience in software development and GUI design
- Furthered my understanding of distributed computing and parallel programming
- Improved my problem-solving and research skills
- Confirmed my interest in pursuing a career in research and development

Code available at:

<https://github.com/Lucas-Labouret/BlobComputing>

Thank you for your attention!

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